Autonomous reaction-energy recycling: complete kinetic constructions and the unresolved physical endpoint

Research record • 4 November 2026 • Independent construction, adversarial comparison, and reproducible calculations

Status and principal result

The requested civilisation-changing mechanism or capability has not been established. This record preserves substantial completed conditional science without presenting it as that endpoint. The selected question concerns autonomous catalytic energy recycling: can energy released by one reaction be captured and routed to another reaction so that the complete chemical cycle becomes predictably faster, after thermal excitation, reverse reactions, readiness, relaxation, transport, and measurement uncertainty are included?

The most useful completed construction is an exactly solved, thermodynamically consistent energy relay with matched capture-on and capture-off comparators. It proves that a relay can have exactly thermal conditional excitation populations, zero net dissipation through the excitation bath, and a causal turnover advantage. Its advantage comes from removing a kinetic deficit of the capture-off process. Therefore a bound on the operating relay's departure from thermal equilibrium cannot alone bound the entire benefit of energy routing. Both physical comparators must be specified.

That construction also quantifies its price. With a hypothetical 100-second excitation lifetime, its gain is 9,855.74, but it has multi-hour stalls and a large current variance; one isolated unit needs approximately 8.38 years for a 1% asymptotic relative standard error. At a 41-nanosecond lifetime and the same chemical rates, its gain is only 1.00000404. These are predictions of a specified synthetic model, not measured catalyst performance.

Other completed results connect finite pulses to signed net turnovers, repair a reversible-kinetics failure in a simple pulse argument, provide a computable thermal-projection certificate for finite reaction networks, and solve a reversible 72-state two-site catalytic model. The physical generator, excitation law, and matched material intervention required to explain the observed PtCo/calixarene enhancement have not been acquired. The distinction between a finite mathematical construction and that unresolved physical operation is maintained throughout.

1. Independent questions and selection

Three constructors began with independent authority over their questions and domains. They did not start as literature, proof, and implementation workers inside a common architecture. The first-principles constructor recorded its mechanism and deductions before searching for nearest prior art. Its first checkpoint is PRE_LITERATURE_20261104T215417Z.txt; its second is SECOND_PRE_LITERATURE.txt. The formal selection record was written later and is explicitly not a backdated preregistration.

The observations route initially asked whether the recent atmospheric methane acceleration could be attributed more sharply to changing microbial emissions or changing hydroxyl removal by combining concentration and two isotopic observables. It reconstructed public annual data and source tables, then performed the proposed inverse operation. Several apparently conflicting percentages referred to different time windows and denominators. A well-mixed isotope contrast canceled hydroxyl only under restrictive mixing and fractionation assumptions; spatially selective removal reintroduced an uncontrolled term. The dual-isotope inverse was algebraically identifiable with fixed signatures, but public Monte Carlo covariance made the source-sink contrast statistically weak. A favorable generalized least-squares design gave an OH uncertainty of roughly 14.8–16.1 Tg/year for a comparison differing by about 14.3 Tg/year. No new atmospheric mechanism or decisive attribution was earned. The calculation, covariance inputs, source hashes, and repair are preserved.

The transfer route asked whether tools from stochastic control, electrical capacity, and certified numerical analysis could turn controlled local observations into reliable rare-event predictions without waiting for spontaneous barrier crossing. It constructed finite-time force acquisition, repaired a cut-locus error in circular increments, included sensor error and diffusion estimation, and built exact finite-element capacity brackets. It then challenged the crucial final operation: capacity divided into basin mass is not generally the requested mean first-passage time from an arbitrary preparation. Even reflecting Brownian motion on [0,1] separates the two quantities. A mixed Poisson formulation gives the correct target, but conditioning, geometry, and acquisition costs remain and may be enormous. The result is useful conditional analysis, not a general efficient rare-event capability. The capacity theorem, exact rational examples, controlled-observation interface, and governing Astra examples are preserved.

The first-principles route first invented a representation that retains only the future-relevant boundary of a structured stochastic generator, deriving a reduction in unnecessary memory erasure and a finite-block overhead. Close prior art already supplies the central thermodynamic construction. It then asked whether repeated generative retraining can be understood as a probability-valued martingale whose variance is consumed by finite resampling. It derived collapse and replenishment recurrences before checking the literature. Existing primary work already establishes the central phenomenon and relevant martingale mechanisms. These were not renamed as new principles. Both pre-literature deductions and exact prior-art comparisons remain in the package.

The observations constructor then developed a materially different question around experimentally reported vibration-coupled catalysis. It attempted the missing pulse-to-product calculation rather than merely proposing an experiment. This opening was selected because a consequential physical claim, a public supporting dataset, and a specific missing operation could be brought into direct contact. The intended endpoint remained predictive autonomous chemical-energy routing, not the smaller theorem eventually proved. Only after the empirical question was anchored did the root mine the relevant Astra catalytic sources. The constructors then challenged one another's accounts, producing the reversible escape, the full network, the thermal-projection bound, and finally the matched relay that limits the bound's interpretation.

2. Empirical anchor and what existing accounts already explain

Liang and colleagues report PtCo catalysts modified with tert-butylsulfonylcalix[4]arene and attribute improved oxygen reduction to vibration-coupled energy transfer [1]. Their proposed causal chain is O2 adsorption, excitation of anchored flexible calixarene, phonon transfer to a neighboring OH* site, and easier OH*-to-water conversion/removal. The public abstract gives 0.126 eV per molecule, a calculated barrier change from 0.375 to 0.183 eV, 13.2% adsorption-energy utilization, mass activity 1.03 A/mgPt, and peak fuel-cell power 2.07 W/cm² at 0.1 mgPt/cm². The exact public supporting information was downloaded and read, including rendered plots. The full main article was not acquired; absent access is not evidence that a method or control is absent from the article.

The supporting information annotates a 41 ns storage time. Its derivation was not acquired. The 0.48 ps propagation estimate is a distance divided by sound speed, not a measured excitation relaxation time. The 0.126 eV figure is a single-point molecular electronic-energy increase, not an experimentally established two-state free-energy gap. Treating it as a Boltzmann excitation gap is an explicit modeling assumption in our examples. The reported barrier reduction also does not by itself specify its duration, operating-potential dependence, prefactor change, or arrival distribution at target sites.

The source's Figure S16 has a U-shaped TOF ratio of approximately 3.14–3.49 across 260–340 mV. Our digitization uncertainty, about 0.009 in ratio for a 1.5-pixel reading error, is image-extraction uncertainty, not an experimental confidence interval. The separately fitted Tafel slopes, 66.1 ± 0.3 and 60.2 ± 0.2 mV/decade, use different narrow fitting windows. They cannot be extrapolated as globally valid slopes over the complete ratio curve.

Normalization and material changes matter. The reported O2 uptake used in site counting changes from 0.079 to 0.04175 mmol/g, a ratio of 0.52848. Under the corresponding same-mass mapping, a 3.3 TOF gain implies a current gain of 1.74399. This does not make the enhancement an artifact: site blocking, accessibility, and probe stoichiometry are possible physical explanations of the denominator. It does mean that intrinsic-site and mass-normalized gains are different observables. The supporting information also reports structural changes and a 35% reduction in a pressure-independent oxygen transport resistance. A matched causal comparison must retain those alternatives rather than treating all performance changes as a vibrational measurement.

Strong existing ORR treatments already explain much of the ordinary behavior. Hansen, Viswanathan, and Nørskov's microkinetic account connects OH coverage, kinetic and thermodynamic volcanoes, an approximately 59 mV/decade Tafel regime, and first-order oxygen dependence under its stated assumptions [2]. A uniform 3.3-fold elementary-rate change at 298 K corresponds to only kBT log(3.3) = 0.03066 eV; this is an equivalent Arrhenius shift, not a claim that the whole catalytic cycle is controlled by one step. Static free-energy changes, site populations, solvent environment, and transport therefore remain quantitatively plausible competitors.

Autonomous energetic coupling is also not a new idea introduced here. Decarolis and colleagues provide operando evidence for coupled CO oxidation, carbon deposition, and carbon removal in Rh catalysts, including spatial spectroscopy and dilution interventions [3]. Their published correction is included in the source record. This is an independent-condition confirmation that reaction coupling can be physically relevant, not a new prediction of our

framework and not validation of this PtCo mechanism. Dynamic gating and fluctuating catalytic pathways likewise have established primary theory [4].

The actual scientific question is consequently narrower and stronger than whether energy can couple reactions: can the proposed capture and transfer explain the measured voltage-dependent complete-cycle enhancement after all known competing accounts and kinetic costs are retained, and can the resulting account predict a new intervention quantitatively?

3. From a short pulse to a complete turnover

Start with an explicitly conditional waveform. Suppose a barrier reduction decays as DeltaE(t) = DeltaE0 exp(-t/tau), the rate prefactor stays fixed, and transition-state kinetics applies. Write c = DeltaE0/(kBT). The rate multiplier is u(t) = exp[c exp(-t/tau)]. Its integrated excess multiplier is exactly

```
I_exp = integral_0^infinity [u(t)-1] dt
= tau [Ei(c) - EulerGamma - log(c)].
```

For DeltaE0 = 0.192 eV, T = 298 K, and tau = 41 ns, I_exp = 11.5311 microseconds. A maximum-height rectangular envelope with the same duration gives I_rect = tau[exp(c)-1] = 72.3841 microseconds. The rectangular number can also bound a fixed-height pulse with a specified mean lifetime; it is not the value for every conceivable waveform. These exposures are not lifetimes and are not themselves extra products.

For an irreversible serial cycle with cold rates k_i and actual conditional rates k_i u_i(t), let X be the current chemical state. Under stationary flow and finite occupations,

```
J0 = 1 / sum_i(1/k_i),
J/J0 = 1 + sum_i <(u_i-1) 1{X=i}>.
```

This is exact even when the multipliers depend on the past. Each chemical transition has mean flux J, and summing its occupancy J/k_i minus its excess conditional hazard gives the identity. It accounts for the fact that speeding one elementary step does not eliminate time spent in all other steps.

If received pulse envelopes are bounded by an integrable nonnegative envelope of exposure I, overlapping pulses are non-superadditive under the stated envelope rule, and the received-event rate is nu, then J/J0 <= 1 + nu I. If the productive donor budget also guarantees nu <= mJ, the sustained gain G satisfies

```
G <= 1/(1 - m J0 I), provided m J0 I < 1.
```

No Poisson arrival assumption, independent branching approximation, or perfect phase mixing is needed for this upper bound. It already allows favorable pulse timing. However, the productive-donor assumption is real: gross reversible adsorption attempts need not be bounded by net productive turnovers.

For G = 3.3, the necessary received-event rates are 199,460/s for the exponential waveform and 31,775/s for the rectangular envelope. At J0 = 4/s, the necessary productive donor budgets are m >= 15,111 and m >= 2,407. At J0 = 0.2/s they become approximately 302,212 and 48,144. With m = 100 and J0 = 4/s, the gain ceilings are 1.00463 and 1.02982. To permit 3.3 at that same donor budget requires lifetimes of at least 6.195 microseconds and 0.987 microseconds, respectively. These optimistic necessary conditions follow from the rate ratio and complete-cycle timing, so the unknown cold attempt frequency cancels within this model class.

These numbers do not exclude the source authors' mechanism. We have not acquired the material's gross reversible activity, pulse waveform, barrier-potential law, correlated readiness, or donor graph. An earlier calculation counted the probability of an added elementary hazard clock and suggested a much smaller target count. That is not a general net-turnover result and was superseded by the complete-cycle ledger. Its original calculation is retained for provenance, with the correction made explicit.

A reversible escape that defeats a tempting generalization

Consider two chemical states E and I with two distinct reversible channels. Precursor formation E -> I has rate c0 = 1000/s and reverse d = 10^6/s. Product formation I -> E has rate a = 1046.4659/s and reverse b = (c0 a/d) exp[-1.2 eV/(kBT)]. The cycle has a large 1.2 eV affinity, yet its stationary productive current is only 1.04433/s. Symmetrically scaling the product channel has a degree of rate control of 0.998956, even though J0/a is approximately 0.000998.

The rare reactive state is continuously replenished and rapidly lost by precursor recrossing. Therefore a large total affinity does not justify replacing a reversible network with serial waiting fractions. The exact finite-pulse signed product response, including all later relaxation, is 0.0000119582 extra turnovers for an unconditioned stationary exponential pulse, but 0.0118794 if the same pulse begins in state I. The rectangular counterparts are 0.0000727621 and 0.0714169. Readiness changes the response by about a thousandfold with no change in pulse height or duration.

This is a prescribed-pulse counterexample, not a free physical state selector. An autonomous catalyst must generate that correlation through explicit dynamics and pay the relevant reverse processes and chemical work. It identifies the missing empirical operation: the joint law of donor events, target chemical state, excitation phase, and subsequent signed productive response.

4. A general signed response and a closed reversible network

For a finite baseline continuous-time Markov generator Q0, retain each reaction channel and its signed product increment d_ij. Let r0(i) = sum_j q0_ij d_ij and J0 be its stationary mean. Solve the Poisson equation

```
Q0 h = J0 - r0.
```

For a finite perturbation followed by complete relaxation, the total extra expected product is

```
R = integral sum_ij p_i(t) delta_q_ij(t)
[d_ij + h(j) - h(i)] dt.
```

The stationary analogue uses the corresponding stationary average. The correction h(j)-h(i) charges the later consequences of occupying the destination instead of the source. It can cancel an apparently large positive local contribution, and it can have either sign. This is established Markov reward and performance-difference machinery, applied here to close the missing operation; it is not a new law of catalysis.

The constructors implemented a 72-state model: two six-stage catalytic sites and a binary excitation mode. The chemical sequence is mass-balanced, all reverse channels obey local detailed balance, excitation has a thermal reverse transition, and adsorption capture replaces its ordinary channel rather than silently adding free energy. Separate variants let the hot mode modulate a chemical rate or be consumed by an extra reactive channel. Stationarity, chemical current, energy bookkeeping, total entropy production, and zero-affinity equilibrium were

checked at 55-digit precision. The supplied free energies and rates are synthetic assumptions, not a fit to the material.

With other forward chemical rates of 100/s and a 41 ns mode lifetime, a consumption variant gives pair currents 3.60827270279/s without capture and 3.60827272038/s with capture. The net change is only 1.7593e-8/s. Its Poisson ledger contains a capture contribution of about +3.23971291445/s and an ordinary-pathway replacement contribution of about -3.23971289685/s. Omitting the negative term would produce a spurious catalytic enhancement.

At other forward rates of 10^6/s the same construction gives a capture gain of about 1.02364. Extending the lifetime changes the regime: at 0.41, 4.1, and 41 microseconds, one variant gives gains of 1.27762, 3.52881, and 9.20713. The approximately threefold regime requires a lifetime 100 times the source annotation and currents thousands of turnovers per site per second. It cannot be transferred to the reported slow regime without changing the model. A separate weak-binding stress case makes gross adsorption exceed net turnover by over a millionfold and produces a small slowdown, ensuring that the donor-budget assumption was not silently restored.

An exact renewal calculation independently treats a six-step forward cycle and finite gate switching. It propagates the gate distribution through each chemical phase with H_i = (D_i-G)^(-1) R_i and residence vector t_i = (D_i-G)^(-1) 1. This preserves readiness at arrival rather than replacing it by a global excitation fraction. For one J0 = 1/s benchmark, a pulse at the same site's adsorption has negligible effect by the time the site reaches OH*, while a perfectly aligned pulse produces gain 1.00007238. This does not rebut the source's neighboring-site mechanism; it quantifies why the timing and spatial architecture cannot be omitted.

5. A finite thermal-projection certificate, with its exact scope

The full network suggests a useful mathematical operation: bound the effect of finite nonequilibrium excitation without pretending that a mean stored energy determines the current. Let i index chemical configurations and g = 0,1 a gate. The undriven bath excites the gate at gamma_i q_i and relaxes it at gamma_i, where q_i is the thermal rate ratio. Its conditional equilibrium hot probability is t_i = q_i/(1+q_i). All other reaction and capture channels are retained, including reverse channels and product rewards.

Let K_g be the chemical generator conditional on gate state and r_g the instantaneous signed product reward. Construct a comparator by thermally averaging each row:

```
Kbar = (1-t) K0 + t K1; rbar = (1-t) r0 + t r1.
mubar Kbar = 0; Jth = mubar rbar.
Kbar h = Jth - rbar.
w_i = r1(i)-r0(i) + [(K1-K0)h]_i.
```

This comparator preserves the model's chemical architecture and uses instantaneous conditional bath equilibration. It is not generally the bare catalyst and is not generally the capture-off process. If mu_i is the actual chemical occupancy and p_i the actual conditional hot probability, stationarity gives the exact identity

```
J - Jth = sum_i mu_i (p_i-t_i) w_i.
```

The bath contribution to entropy production, measured in kB per unit time, is

```
sigma_gate = sum_i gamma_i(1+q_i) mu_i
[D_Bern(p_i || t_i) + D_Bern(t_i || p_i)]
<= sigma_total.
```

Pinsker's inequality and Cauchy–Schwarz then give

```
|J-Jth|² <= sigma_total V_mu <= sigma_total Vmax,
V_mu = sum_i mu_i w_i²/[4 gamma_i(1+q_i)],
Vmax = max_i w_i²/[4 gamma_i(1+q_i)].
```

This is a full-network coefficient: w includes all downstream chemical consequences, not merely the hot-minus-cold elementary rate. If there is one thermodynamic affinity A and the counted product is its conjugate cycle current, sigma_total = A J. For positive Jth, define kappa = A Vmax/Jth and

```
B(kappa) = 1 + kappa/2 + sqrt(kappa + kappa²/4).
1/B(kappa) <= J/Jth <= B(kappa).
```

Multiple donor fuels require their complete sum of affinity times current. One cannot identify donor current with target product current by convenience. For two physical models indexed by 0 and 1, the valid comparison is

```
(Jth_1/Jth_0)/(B1 B0) <= J1/J0
<= (Jth_1/Jth_0) B1 B0.
```

This retains changes in thermally projected pathways and the capture-off deficit. Both generators and both physical resource ledgers must be supplied.

Completion beyond a binary gate

For a supplied finite irreducible bath generator B with equilibrium distribution nu and a nu-centered corrected observable w, solve Bf = w. A current identity and the logarithmic mean L(a,b) = (a-b)/(log a-log b) give

```
|E_p w|² <= D_B(p) sum_(x<y)
L(p_x b_xy, p_y b_yx) [f(y)-f(x)]².
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Because L(a,b) <= (a+b)/2, a directly computable uniform coefficient is

```
C = (1/2) max_x sum_y b_xy [f(y)-f(x)]².
```

A sharper coefficient maximizes the concave logarithmic-mean expression over the probability simplex. This closes the finite mathematical operation through stationary solves, Poisson solves, and finite optimization rather than assuming access to an unknown functional-inequality constant. With N supplied chemical states and at most m gate states per chemical state, dense linear algebra costs $O(N^3 + N m^3)$, plus channel aggregation. N can itself grow exponentially with interacting sites; physical state discovery and rate acquisition are not included in this arithmetic cost.

For binary gates, the constant 1/4 improves to the exact secant supremum c(t) = sup_p (p-t)/[logit(p)-logit(t)]. For the illustrative gap 0.126 eV at 298 K, t = 0.0073432 and c(t) = 0.12688335. A further exact-cost dual bounds J with a one-dimensional convex inner optimization. In the 100/s 72-state example, the actual capture-on ratio J/Jth is 0.99999112. The simple, secant, and numerically optimized dual upper bounds are respectively 1.18840, 1.13093, and 1.03018. These numerical optimizations are not outward-rounded interval

certificates. At other rates of 10^6/s the same bounds are loose, and those unsuccessful cases are preserved.

The independent challenge also constructed an asymptotically sharp four-state family for the absolute inequality. It is a slowdown, so it does not establish sharpness of the positive-gain bound. A separate exact six-state model gives J/Jth = 3.5217684. Its two productive cycles compete for a shared state: finite gate switching reduces occupation of a slow branch from about 98.89% to 63.67%. This is ordinary kinetic trapping and cycle selection. Its maximum-based upper bound is 4515.77, which is valid but weak. Another exact example has gain 1.01987 and bound 1.60206, showing conditional nonvacuity without universal usefulness. For a simple reversible single ring, the matrix-tree expression makes current monotone in symmetric bath-edge rate; the positive example needs more topology than that restricted model.

What the strongest existing mathematics already supplies

The entropy cost of holding a system away from its undriven equilibrium is established: Horowitz, Zhou, and England explicitly derive the undriven-edge maintenance cost and its fast-control limit [5]. General observable response and relative-entropy inequalities are also established [6]. Aslyamov, Ptaszyński, and Esposito derive exact fluctuation-response relations and partial-entropy bounds for Markov currents [7]. Nagayama, Yoshimura, and Ito supply the general-current/logarithmic-mean entropy machinery used in the finite-gate completion [8]. Conditional-generator coarse graining and finite kinetic sensitivity are existing topics [9,10].

The precise composition above is a useful finite certificate with a carefully chosen comparator. No identical theorem was established in the targeted sources checked, but that is not an exhaustive novelty result. The ingredients are familiar, and no new foundational thermodynamic principle is claimed. Dissipation-only rate bounds for external forcing cannot simply be imported: the governing Kuznets-Speck–Limmer construction assumes forcing dynamics independent of the system state, excluding autonomous feedback, and additional activity terms or limiting conditions matter [11].

6. The exact relay that repairs the causal interpretation

The strongest constructive challenge uses four states labeled by chemical configuration and gate, 00, 01, 10, 11. Their free energies are 0, E, -E, 0. Define q = exp[-E/(kBT)] and z = exp(-A), where A is the dimensionless chemical affinity. In each chemical configuration, the bath flips the gate upward at gamma q and downward at gamma.

The capture-on model has two distinct channels between 00 and 11. Adsorption capture is 00 -> 11 at rate a and its reverse at rate a. Product formation is 11 -> 00 at rate c and its reverse at rate cz; the product channel counts +1 or -1. Keep these channels separate even though their endpoints coincide: combining them would erase the chemical work and energy ledger. Capture stores E in the gate while lowering chemical free energy by E. The product channel consumes the mode and closes the chemical cycle. Every reverse is present and obeys local detailed balance.

The matched capture-off model retains the product channel and bath. It replaces capture 00 <-> 11 with ordinary adsorption 00 <-> 10 at rates a and aq. The two models have the same state free energies, chemical affinity, and thermally projected chemical generator Kbar. This is therefore a stronger comparison than selecting an arbitrarily slow unrelated baseline.

The stationary solution is elementary. Set t = (a+cz)/(a+c). In the on model,

```
pi00 = 1/[1+q+t(1+1/q)],
pi01 = q pi00; pi11 = t pi00; pi10 = pi11/q.
```

Both conditional gate distributions are exactly thermal. Net bath heat and bath entropy production vanish, while the mean captured and consumed excitation-energy flux is EJ and total entropy production is AJ. Define

```
D0 = (1+q)(1+z/q)/a + (1+q)(1+1/q)/c,
C0 = 1+q+1/q+z.
J_on = (1-z)/D0 = Jth,
J_off = (1-z)/(D0+C0/gamma),
J_on/J_off = 1 + C0/(gamma D0).
```

The relay has a causal gain despite J_on-Jth = 0. It removes the off model's finite-relaxation deficit. This is why the single-model thermal-projection inequality cannot be advertised as a bound on all energy-recycling benefit.

Mean energy balance does not establish the ancestry of a product's excitation. To track ancestry, label each entry into 11 by capture, thermal excitation, or reverse product, and clear the label on every exit. The fraction of forward products whose immediately preceding entry was capture is

```
f_capture = pi00 a /
[pi00(a+cz) + pi10 gamma q].
```

This is an explicit trajectory property of the specified memoryless model, not a claim that an indistinguishable physical quantum carries an observable serial number. It separates direct retention from repeated bath replacement.

For E = 0.126 eV, T = 298 K, a = c = 100/s, and A = 1.2 eV/(kBT), the exact predictions are:

```
Lifetime 41 ns: J_on = 0.723615469/s;
J_off = 0.723612545/s; gain = 1.00000404044;
f_capture = 0.00000819993; cold-stall mean = 5.542 microseconds.

Lifetime 100 s: J_on = 0.723615469/s;
J_off = 0.0000734207/s; gain = 9855.74161;
f_capture = 0.999950003; cold-stall mean = 13,518 s.
```

The current fluctuations were also solved, rather than inferred from a Poisson approximation. If Qh = J-r, the asymptotic variance rate is the channel sum of stationary forward-plus-reverse traffic times $[d+h(j)-h(i)]^2$. Its ratio to J, the Fano factor, is 0.98564 in the fast-bath case and 19,137.90 in the slow-bath case. A single unit needs asymptotic observation time T = variance_rate/(epsilon^2 J^2), giving approximately 13,621 s and 264,476,084 s for epsilon = 0.01. These are standard-error estimates after mixing, not finite-time confidence guarantees. Independent parallel units reduce time in proportion to their number only when independence and equivalent acquisition are actually secured.

The 100-second lifetime is hypothetical, roughly 2.44 billion times 41 ns. The model's product is a lumped two-stage chemical cycle, not a literal elementary four-electron ORR mechanism. The exact gain is therefore an existence construction and a resource calculation, not a proposed economical fuel-cell catalyst or an explanation of the observed material. It illustrates why very large fold gains can coexist with modest absolute throughput and severe intermittency.

7. Predictions, confirmations, and the remaining physical operation

The following consequences are new deductions of the specified constructions in this run, not empirical discoveries. Within the serial productive-donor class, finite pulse exposure imposes the quantitative donor-rate and lifetime requirements above. Within the reversible two-channel example, conditioning on the rare reactive state changes the complete pulse response by roughly a thousandfold. Within the exact matched relay, gain is linear in inverse bath rate at fixed a, c, q, and A, while J_on is independent of that bath rate; excitation ancestry and burst noise change strongly. Within the two-cycle topology, slowing thermal mixing can redirect occupation away from a kinetic trap. These are falsifiable model consequences, but none has been validated on an independently acquired PtCo dataset.

Existing ORR microkinetics and the Rh thermal-levering observations are prior explanations and prior confirmations. They cannot be counted as successful new predictions. Digitizing the same source's gain curve and then reproducing it would be a fit, not an independent test. We deliberately did not fit an unspecified pulse-rate function to the U-shaped curve.

The material-level operation still required is acquisition of the joint distribution of donor events, excitation amplitude and phase, target intermediate, and subsequent signed product response, plus a physical capture-off comparator whose static chemistry and transport changes are measured. An average stored energy, average pulse frequency, and mean lifetime do not determine this joint distribution. The reversible example supplies a concrete counterexample to such a reduction.

The physical resource bill has several distinct parts. The chemical model needs potential-dependent forward and reverse rates or justified intervals, site counting with uncertainty, and a definition of which counted events complete the same net reaction. The excitation model needs a calibrated spectral/state description, its thermal reverse process, actual relaxation waveform, and capture/consumption channels. The spatial model needs the number and distribution of reachable donor-target pairs, distances, correlation, and finite propagation. The execution model needs thermal and mass transport, electrode potential, degradation, preparation, and reset. The comparator needs control of static adsorption and accessibility changes. No economical device capability follows before these are paid.

The precision requirements are consequential. At 298 K, a 5% uncertainty in a single Arrhenius rate corresponds to about 1.25 meV of barrier uncertainty when the prefactor is known; a 10 meV error changes that rate by a factor of about 1.48. These are sensitivity conversions, not claimed measurement capabilities or uncertainties of the source's DFT. In the serial bound G = 1/(1-rho), rho = m J0 I, a target gain of 3.3 gives rho = 0.69697. Approximately 4% relative uncertainty in rho already produces roughly 10% relative uncertainty in the gain under linear propagation. Near a kinetic threshold, rate and topology uncertainties need full correlated propagation.

A 41 ns lifetime has reciprocal scale 24.4 MHz. That gives a dimensional timing requirement for resolving a single exponential relaxation, not a complete instrument design. The source's sub-picosecond propagation estimate concerns a different timescale. Ensemble spectroscopy, electrochemical current, and single-event causal conditioning are not interchangeable measurements. Claiming an executable capability would require an actual calibrated measurement and control interface, not just these reciprocal-frequency estimates.

The mathematical certificate is more general than one chosen catalyst, but its application is conditional on acquired generators. It can compare alternative mechanisms, constrain

improvements from specified gate dynamics, and expose substitutions of gross activity for productive current. It cannot identify an unknown physical mechanism from a mean rate alone. That limitation is the remaining operation separating this branch from its consequential endpoint.

8. Exact Astra use and preservation of the previous branch

Astra was consulted at pinned commit d5ac5437c8f953fc515f931fbcb1ef80b4e96dd7 after the catalytic question was anchored. Navigation entries were not treated as governing theorems. Three exact catalytic sources were read [12]. The causal-identifiability theorem shows that static endpoints and a periodic mean do not bound an isolated pulse's effect or its required conditioning horizon, within an explicit positive switched-system class. The GC proofs distinguish scalar relaxation-mixture monotonicity from sequential multi-state reversals, preserve a poor held-out fit, and state the randomized pulse-omission identification conditions. The activation-cost proof charges unsuccessful attempts, loss, recovery, and overhead in a constant competing-hazard renewal model. Its hypotheses do not automatically apply to an autonomous state-dependent excitation network.

The rare-event route also read the exact ordinary-Langevin mean-passage and well-redistribution sources [13]. Their known homogeneous quartic geometry, mobility restrictions, preparation, and target sets are preserved. Their high-dimensional calculation is not a theorem that arbitrary unknown landscapes admit efficient prediction after free state acquisition. The package records source locations, hashes, and conditions without republishing unlicensed full source collections.

The prior clocked-grid report and archive are preserved byte-for-byte, including the unsuccessful later-week replay. Their SHA256 digests are respectively 4b633e4588c9bdf75cd7e44e605dd9d4b1c457ab0dbb64266f528ec632ec8277 and 3fc271fac70d893748c2cc0d65a13149b2e43b3ab74a70eccd443043ddd0915f. Hartmann and colleagues' source explanation associates the observed oscillations with synchronized UPS self-tests; that is retained as the source authors' causal account, not claimed as our discovery [14].

The mathematical scheduling record retains the exact weighted-midpoint discrepancy e_max, the long-horizon arbitrary bounded per-event timing-error minimax e_max + 2E delta/T, the distinction from fixed device offsets, signed waveform and residual norms, and the stable causal bounded-variation response bound B(|g(0)|+TV(g))/2. The first-order frequency model M f' + D f = -u gives B/M under its stated assumptions. These do not acquire physical AC-grid response coefficients. The frozen week-one offsets [0,12,45] worsened 10 of 15 later weeks, with median ratio 1.0477 and maximum 1.1408. That replay is retrospective, not a field trial. The complete construction, exact waveform classes, limitations, author explanation, and failed replay remain in the unchanged original files.

9. What this run earns

This run changed selection as requested, executed independent scientific constructions, used primary measurement records and exact governing sources, and allowed the constructions to invalidate one another's overbroad interpretations. It did not retain the grid topic because it was familiar. It did not present the quickest closed theorem as the mission endpoint.

The earned output is a complete conditional framework for computing and bounding net catalytic effects of transient energy, with autonomous reversible examples, matched

comparisons, finite algorithms, uncertainty calculations, and explicit failures. The key interpretive result is that thermal marginals and zero net bath dissipation in the operating device do not exclude useful energy routing; the causal comparison can reside in a deficit of the disabled device. This is a concrete counterexample to an overbroad diagnostic, not a claimed new foundational mechanism.

The full PtCo causal attribution remains unresolved, as does a general demonstrated energy-routing capability. No new foundational principle has survived the strongest applicable prior-art challenges in this run. Reporting that boundary is part of preserving the science. The package retains the promising alternatives and the exact remaining operations so that subsequent work can challenge or extend the constructions without rediscovering their failures.

References and governing records

[1] Liang et al., Dynamic Vibration-Coupled Energy Transfer for Boosting ORR Catalysts in Fuel Cells, Journal of the American Chemical Society 148, 5115–5124 (2026). https://doi.org/10.1021/jacs.5c15936 . Public supporting information: https://api.figshare.com/v2/articles/31199023 and https://ndownloader.figshare.com/files/61491559 . SI SHA256: c22504553ca4bb97fbde0dbf23474faf9e2f925181332718d08cb130e921b603 . SI license: CC BY-NC 4.0. Main article: abstract acquired, full text not acquired.

[2] Hansen, Viswanathan, and Nørskov, Unifying Kinetic and Thermodynamic Analysis of 2 e− and 4 e− Reduction of Oxygen on Metal Surfaces, Journal of Physical Chemistry C (2014). https://doi.org/10.1021/jp4100608 . Exact primary model and its coverage, rate, and fitting assumptions govern the comparison.

[3] Decarolis et al., Nature Catalysis 7, 829–837 (2024). https://doi.org/10.1038/s41929-024-01181-w . Full primary account acquired. Correction: https://doi.org/10.1038/s41929-024-01240-2 . Operando and dilution results are prior empirical evidence, not predictions of this report.

[4] Spouge, Szabo, and Weiss, Physical Review E 54, 2248 (1996). https://doi.org/10.1103/PhysRevE.54.2248 . Jangid, Chaudhury, and Kolomeisky, ACS Catalysis (2026). https://doi.org/10.1021/acscatal.6c05102 . Primary dynamic-gating and catalytic-fluctuation comparisons; detailed acquisition notes are in the constructor reports.

[5] Horowitz, Zhou, and England, Minimum energetic cost to maintain a target nonequilibrium state, Physical Review E 95, 042102 (2017). https://doi.org/10.1103/PhysRevE.95.042102 ; https://arxiv.org/abs/1703.04722 . Governing equations 3–4 and 11, with fixed undriven edges and separately counted control channels. Distinct related paper: Horowitz and England, Entropy 19, 333 (2017), https://doi.org/10.3390/e19070333 .

[6] Dechant and Sasa, Fluctuation-response inequality out of equilibrium, PNAS 117, 6430–6436 (2020). https://doi.org/10.1073/pnas.1918386117 . General observable comparison and path-relative-entropy framework.

[7] Aslyamov, Ptaszyński, and Esposito, Nonequilibrium Fluctuation-Response Relations: From Identities to Bounds, Physical Review Letters 134, 157101 (2025). https://arxiv.org/abs/2410.17140v2 . Exact identities 5–8 and partial-entropy response bounds; the derivative conventions matter.

[8] Nagayama, Yoshimura, and Ito, Infinite variety of thermodynamic speed limits with general activities, Physical Review Research 7, 013307 (2025). https://doi.org/10.1103/PhysRevResearch.7.013307 ; https://arxiv.org/abs/2412.20690 . General-current bounds and logarithmic-mean machinery, equations 22, 26, 30, 31 and Appendices E/F.

[9] Hilder and Sharma, SIAM Journal on Mathematical Analysis 56, 913–954 (2024). https://arxiv.org/abs/2201.10256 . Their stationary conditional reference differs from the bath-only reference used here; this difference prevents direct substitution without checking hypotheses.

[10] Owen and Horowitz, Nature Communications 14, 1280 (2023). https://doi.org/10.1038/s41467-023-36705-8 . Positive-observable sensitivity and graph-support bounds. Bebon and Speck, Physical Review Letters 136, 137401 (2026). https://doi.org/10.1103/jcm3-57d8 ; https://arxiv.org/abs/2602.20321 . Tree-polynomial finite kinetic response.

[11] Kuznets-Speck and Limmer, Dissipation bounds the amplification of transition rates far from equilibrium, PNAS 118, e2020863118 (2021). https://doi.org/10.1073/pnas.2020863118 ; https://arxiv.org/abs/2009.04481 . State-independent forcing, rare-event preparation, support, and activity conditions were checked before deciding this is not a general autonomous-feedback bound.

[12] Astra repository: https://github.com/harjassand/astra-research-knowledge/tree/d5ac5437c8f953fc515f931fbcb1ef80b4e96dd7 . Exact catalytic paths at that commit: updates/GA-conformal-causal-genesis-20261016/package/derived/root/CAUSAL_IDENTIFIABILITY_THEOREM.md ; SHA256 3135f96dfaa2f0bf418b9eacf3700dba416d591ca1cec9bfc3724f866bd0e713 . updates/GC-peak-stability-genesis-20261016/package/derived/workers/gc_catalysis_genesis/PROOFS.md ; SHA256 080648041fc39e060d340001c16438f4095ff302d42296340e8cf1db1f116fc8 . updates/FL/synthesis/activation_audit/proofs/activation_bound.md ; SHA256 f3cc65aa6f90bed16b8d24ec2a4fa53fbc44561c132c7a2f2e3e66f4b7e2d674 .

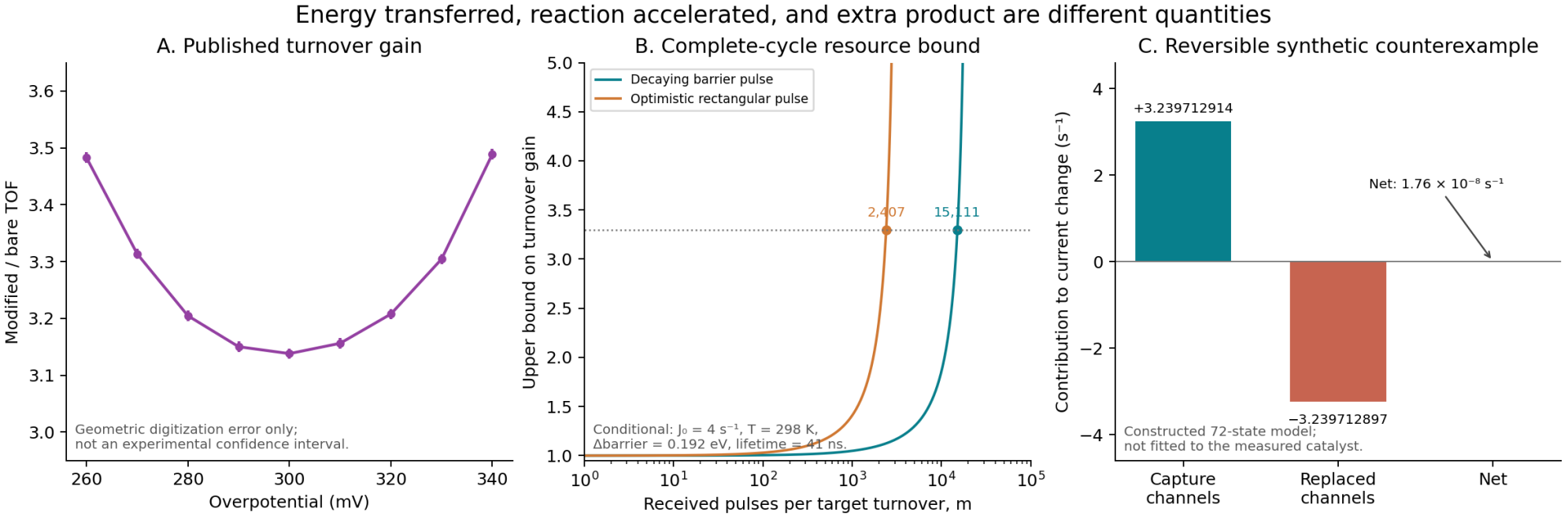
[13] Same pinned Astra commit. Under research/2026-11-01-reviewed-actual-kinetics-retained-clocks-and-task-interfaces/source/dots-new/, governing ordinary-Langevin source: 04-Astra_Ordinary_Langevin_Mean_Passage_Time_20261031/members/audit_constant_spd_mfpt_theorem_20261031_2239/received_candidate.md ; SHA256

bf9e83e1f39f25758dc02b10a06d06a955b98e61de172f633ba5d4f341a7d8e0 . Governing well-redistribution source SHA256 207d9ee314c2c3f4638d0a3c45eaa7e23fd85246010ef799fd538201a0278ab7 ; complete path is in the preserved transfer construction and source manifests.

[14] Hartmann et al., Nature Energy (2026). https://doi.org/10.1038/s41560-026-02144-0 . The original grid report retains the measured events, author explanation, exact theorem conditions, source data, and failed later-week replay.

[15] Preserved alternative primary sources: Oh et al., Nature Communications (2026), https://doi.org/10.1038/s41467-026-74777-4 ; Riddell-Young et al., PNAS (2025), https://doi.org/10.1073/pnas.2516543122 ; measurement-code archive https://doi.org/10.5281/zenodo.17238656 (CC BY 4.0); NOAA public methane series. The first-principles prior-art record includes Loomis and Crutchfield, Physical Review Research 2, 023039 (2020); Garner, Chaos 31, 063131 (2021), https://doi.org/10.1063/5.0044741 ; Borkar, https://arxiv.org/abs/2506.09401 ; He, Xu, and Cheng, https://arxiv.org/abs/2502.18049 ; Gilestro, https://arxiv.org/abs/2609.18560 . Exact deduction and challenge checkpoints are retained rather than overwritten.

Figure: evidence and conditional whole-cycle limits



The left panel digitizes the source supporting information; its error scale represents image extraction, not experimental confidence. The other panels are model calculations under explicit waveform, cycle, and donor-budget assumptions. They are not fits to the material and do not exclude reversible or correlated mechanisms outside that class. Complete scripts and input attribution are included in the accompanying archive.